VITyarthi | Open Source Software

The Open Source Audit

A Capstone Project for OSS NGMC Course

Course: Open Source Software | Unit Coverage: 1 – 5
Max Marks: 100

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1. Introduction

Most of the software that powers the modern world — the servers that run websites, the tools that developers use every day, the operating systems inside supercomputers and smartphones — was not built by a single company and sold for profit. It was built openly, shared freely, and improved collectively. That is the essence of open source.

This project asks you to stop and think about that.

You will choose one real open-source software project — something you can download, read about, and interact with on Linux — and conduct a structured audit of it. Not just 'what does it do,' but why it exists, who built it, what freedoms its license actually gives you, what philosophical values it represents, and how it has shaped the world around it.

Alongside the written report, you will write five shell scripts that demonstrate practical Linux skills. These are not meant to be impressive programs — they are meant to show that you understand how open-source tools work at the command line, and how automation connects to the philosophy of sharing and transparency.

Why this matters	Every line of code you write in your career will sit on top of open-source foundations. Understanding the philosophy behind that is not optional — it is part of being a professional.

2.Choosing Your Software

Pick one open-source project from the list below, or propose your own (subject to instructor approval). Your choice will be the subject of your entire report and all five shell scripts.

Approved software options

Software	Category	License	Why it is interesting
Linux Kernel	Operating System	`GPL v2`	*The foundation everything else runs on*
Apache HTTP Server	Web Server	`Apache 2.0`	*Powers ~30% of all websites globally*
MySQL	Database	`GPL v2 / Commercial`	*A license with a dual-model story to tell*
Firefox	Web Browser	`MPL 2.0`	*A nonprofit fighting for an open web*
VLC Media Player	Multimedia	`LGPL / GPL`	*Plays anything — built by students in Paris*
LibreOffice	Office Suite	`MPL 2.0`	*Born from a community fork — a real lesson*
Python	Programming Language	`PSF License`	*A language shaped entirely by community*
Git	Version Control	`GPL v2`	*The tool Linus built when proprietary failed him*

Your choice	Write your chosen software here before submission: Git_____

3. Project Report Structure

Your report should be 12 to 16 pages long. Each section below maps to a unit in the course. Do not pad with screenshots — every page should contain original thinking and analysis written in your own words.

PartA—Origin and Philosophy(Units 1&2)

This is the most important section of the report. Allocate at least four pages to it.

A1.The problem this software was created to solve

Do not just copy the official description. Research the original context. What was frustrating or broken or inaccessible before this software existed? Why did the creator decide to build it and share it rather than keep it proprietary?

For example: Linus Torvalds was a student who wanted a free Unix-like system and could not afford one. Richard Stallman watched a printer malfunction and could not fix it because the source code was locked. The VLC team wanted to stream video over a university network but notool existed that was free and open enough to adapt.

What is your software's origin story? Tell it properly.

Answer: There was a time during early 2000s, the Linux Kernel was in the middle of a massive logistical nightmare. It was managing millions of lines of code which was contributed by thousands of people which is hard enough, but doing it without a reliable tool is impossible as it caused a lot of problems. At the time, the community was actually using a proprietary tool called BitKeeper. It was a very useful tool out there. It worked well, but there was a catch: the community was only allowed to use it for free as long as they played by the company's rules. The real disaster happened in 2005. When a developer tried to reverse-engineer the BitKeeper protocols to create a free alternative, the company got angry and pulled the plug on the entire Linux community's free license. Suddenly, Linus Torvalds and his team were left in the dark, with no way to coordinate the world's most important open-source project.
When Linus looked at the existing "open" options like SVN or CVS, he found them incredibly frustrating. They were very very painfully slow and and more importantly, they were centralized. We had to be connected to something like a main server just to do basic work. This did not make sense for a global community of hackers. It was giving a feeling like "broken" because it still gave one central point too much power. Torvalds realized that if Linux was going to be truly free, the tool used to build it had to be free too. He did not want to just build another version control system; he wanted to fix the philosophy behind it. He spent a legendary weekend in April 2005 coding the first version of Git. He designed it to be "Distributed" which essentially means that each and every single person working on the project has a full backup of the entire history. This was an excellent move because it gave meaning that no company could ever hold the code "hostage" again, which means no company could never ever again stop a corporate license. By releasing it under the GPL v2 license, he automatically turned Git into a public utility for everyone. It wasn't about making money; it was about making sure that collaboration could never be shut down by a corporate license again. This was the history of a new age software in the making.

A2.The license—what it actually says

Download or find the actual license text for your software. Read it. Then answer these questions in your report:

- What are the four freedoms of free software? Does your software's license grant all four?
- Can a company take this software, modify it, and sell it without sharing their changes? Why or why not?
- What is the difference between GPL and MIT licensing? Which would you choose for something you built, and why?
- Is there a difference between 'free as in free beer' and 'free as in freedom'? Give an example from your chosen software.

Answer: a) For this audit, I mainly looked at the official license for **Git**, which is the **GNUGeneral Public License, version 2 (GPLv2)**. Reading through a official and legal document can be dry, but it is the only way to truly understand what "Open Source" means in a practical sense. **The Four Freedoms.** The concept of Free Software is built on four essential freedoms: these are the freedoms to run the program for any purpose, the freedom to study how it works (which requires access to the source code), the freedom to redistribute copies, and the freedom to improve the software and share those improvements with the public. After reviewing the GPLv2, it is very clear that Git grants all four of these. It does not just let me use the tool, it invites me to look at the C code and shell scripts that make it run, which is incredibly helpful for someone learning about systems and security.

b) This is where the GPLv2 gets interesting. A company can take Git, modify it, and even sell it but they cannot keep their changes a secret even if they distribute that version. This is because the GPL is a "Copyleft" license. It has a reciprocity clause: if you share a modified version of the software, you are legally responsible for sharing the source code of your changes under that same license. You cannot "close" the loop. This is why we do not see a Pro version of Git that has secret, better features hidden away from the community.

c) The main difference is about control. The **MIT license** is "permissive" i.e. it basically says do what you want to but just do not sue me. A company could take MIT code, put it in a private product and never give anything back. The **GPL** however is protective, it ensures the code stays free forever. If I was building a tool today, I would highly like to choose the **GPL** because of mainly student reasons. As a student, I put a lot of work into my projects, and I had want to make sure that if a big corporation uses my work, they have to contribute back to the community rather than just profiting from my effort in secret.

d) There is a huge difference here. "Free as in free beer" just means you did not pay any money for it. "Free as in freedom" means you have the right to control the tool.

- **Example:** If I download a free game on my phone, that is "free beer" I did not pay, but I cannot see the code or change how the game works.
- **Git Example:** Git is both. I did not pay for it, but more importantly if I do not like how a certain Git command works, I have the **freedom** to rewrite the logic and compile my own version. That is really power.

Think, do not copy	Answers copied from Wikipedia or AI tools will score zero. We are looking for your understanding and your words.

A3.The ethics of open source

Open source is not just a licensing model — it is a statement about how knowledge and tools should be shared. Reflect on the following:

- Should all software be open source? Make an argument for and against.
- Is it ethical for a company to build a profitable product entirely on top of free community labor? (Think about how companies like Red Hat, Google, and Meta use Linux.)

- What does 'standing on the shoulders of giants' mean in the context of software? Does open source enable or undermine original innovation?

There are no correct answers here. We are grading the depth of your thinking, not your conclusion.

Answer: a) The argument that I would make for making everything open source is that it accelerates human progress. If every piece of medical software or security infrastructure was open, we could fix bugs faster and save lives. It prevents a monopoly where one company can charge whatever they want because only they know how the code works.
However, the argument against it is mostly about survival and innovation. If a small team thinks to spends five years and millions of dollars developing a very niche and normal specialized tool like a high end 3D engine or a new encryption algorithm they need a way to reinstall that investment. If they are forced to give the "secret sauce" away on day one, they might never build it in the first place. I think a world without any proprietary software might actually slow down innovation in highly expensive, specialized fields.

b) When I see a company like Google or Meta making billions while using the Linux Kernel for free, it feels a bit exploitative at first glance. It is "free labor" for them. But as I have looked deeper into the Git and Linux communities, I have realized it is more of a symbiotic relationship.
Companies like Red Hat do not just take code, they employ hundreds of developers whose fulltime job is to submit patches back to the Linux Kernel. In my view, it starts becoming unethical only when a company takes from the community but contributes nothing not even bug reports or documentation back to the source. If they are active participants in the ecosystem, their profit is actually what helps keep the ecosystem funded.

c) In the context of software, this means that I do not have to start from scratch or evry first line every time I want to build something. When I write a script in Kali Linux, I am actually standing on the work of the people who built the Bash shell, the people who wrote the C libraries, and the people who built the Kernel itself.
I believe open source enables original innovation rather than undermining it. Before Git existed, developers were wasting a huge amount of time just trying to figure out how to share code without breaking it. Because Linus Torvalds shared Git with the world, my generation of developers can skip the how do we share code problem and move straight to solving of how do we build better security tools. Open source provides a high stage to everyone to stand on, so we can reach for an even higher ceiling.

PartB —Linux Footprint(Unit2)

Research and document how your chosen software lives on a Linux system. Where does it install? What user does it run as? How is it managed? This section requires you to actually install the software on Linux — either on your machine, a virtual machine, or a lab system.

- Installation method — is it available via RPM, apt, or must it be compiled from source?
- Key directories — where are its binaries, config files, and logs stored? (e.g., /usr/bin, /etc, /var/log)
- User and group — what Linux user and group does it run under? Why does this matter for security?
- Service management — can it be started, stopped, or restarted as a service? Show the commands.
- Update model — how does the open-source community push security patches to your machine?

Answer: a) For this section, I performed a live audit of Git on my Kali Linux **(**WSL2**)** environment. Like understanding how a tool actually sits on the disk is the very first step in analyzing security, which is why I did not just read the manual I actually checked the system paths myself.

```
┌──(xcenz㉿s7-pm)-[~]
└─$ sudo apt install git
[sudo] password for xcenz:
git is already the newest version (1:2.51.0-1).
The following packages were automatically installed and are no longer required:
  g++-14  g++-14-x86-64-linux-gnu  gccgo-14  gccgo-14-x86-64-linux-gnu  libgo-14-dev  libgo23  libstdc++-14-dev
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1

┌──(xcenz㉿s7-pm)-[~]
└─$ |
```

b) On my Kali system, Git is managed through the apt (Advanced Package Tool) repository. While I could have compiled it normally from the source code available on GitHub, but rather using sudo apt install git is the standard way to ensure, I get a version that is stable enough and pre-tested for my distribution.

Key Directories and Locations I ran a few commands to see where Git stays or lives on my machine.

- **Binaries:** By running which git, I confirmed that the executable is located in /usr/bin/git.
- **Configuration Files:** Git uses a tiered config system. The system wide settings are in /etc/gitconfig, but my personal user settings (like my name and email) are stored in a hidden file in my home directory: ~/.gitconfig.
- **Documentation:** I found the local manual pages stored in /usr/share/man, which I can access anytime using the man git command.

```
┌──(xcenz㉿s7-pm)-[~]
└─$ which git
/usr/bin/git

┌──(xcenz㉿s7-pm)-[~]
└─$ |
```

c) One thing I noticed is that Git does not run as a background root service. Instead of that, it runs under my current Linux user account (xcenz).

Security Significance: This is a critical security feature. Because it runs as a standard user, if I accidentally download a malicious repository, the blast radius of it is limited. The Git process does not have the permissions to delete my system files or modify the kernel unless I explicitly use the sudo method, which follows the Principle of Least Privilege.

```
┌──(xcenz㉿s7-pm)-[~]
└─$ git config --list --show-origin
file:/home/xcenz/.gitconfig     user.name=Ayush Mishra
file:/home/xcenz/.gitconfig     user.email=ayush.24bce11111@vitbhopal.ac.in
```

d) Unlike a web server , Git is a "command-line utility." It does not stay running in the background. Therefore, I do not use systemctl start git. Instead of that, I manage my interaction with it through direct commands. However, if I was using a Git hosting service like **GitLab** on this machine, I would have used:

- sudo systemctl start gitlab
- sudo systemctl status gitlab

e) Security in the open-source world is about speed. When a vulnerability/open thread is found in Git, the core maintainers push a patch to the source code. The Kali Linux maintainers then package that patch and push it to their official mirrors. I receive these updates by running: sudo apt update & sudo apt upgrade. This community pipeline is much more transparent and opaque than waiting for a proprietary company to release a Service Pack.

```
┌──(xcenz㉿s7-pm)-[~]
└─$ ls -l /usr/bin/git
-rwxr-xr-x 1 root root 4155600 Aug 26  2025 /usr/bin/git
```

PartC —The FOSS Ecosystem (Units 3&4)

No open-source project exists in isolation. Map the ecosystem around your chosen software.

- What other open-source tools does it depend on? (Libraries, compilers, runtimes) •
  What has it enabled or inspired? (Forks, spin-offs, tools built on top of it)
- Where does it fit in the LAMP stack, or how does it relate to the web?
- Is there a community — mailing lists, forums, conferences, IRC channels? Who governs it?

Answer: a) No software is an island, and Git is a perfect example of this. When I looked into its source code and installation requirements, I realized it does not just "exist" on its own—it relies heavily on other open-source giants. For instance, it uses zlib for its impressive data compression and libcurl to handle network transfers when we run a git clone over HTTP. It is also primarily written in C, which connects it directly to the GNU Compiler Collection (GCC). Seeing these layers made me realize that even a "foundation" tool like Git is standing on the shoulders of other community-built libraries.

b) While Git isn't a direct "layer" of the LAMP stack (Linux, Apache, MySQL, PHP/Python) like a database or a web server is, it acts as the "glue" that holds the development process together. In my experience, you do not just "deploy" a LAMP application anymore; you version control it with Git first. It has enabled a workflow where open-source web servers like Apache can be updated safely because the code was managed in a Git repository before going live.

c) The beauty of Git is license is that it encouraged others to build on top of it. The most obvious examples are platforms like GitHub and GitLab. Even though GitHub is a proprietary service, the core technology is Git. I also found it interesting that Git has inspired "spin-offs" or specialized versions, like Git LFS (Large File Storage), which solves the specific problem of handling huge binary files that the original Git wasn't designed for.

d) Who "owns" Git? The answer is both "no one" and "everyone". While Linus Torvalds started it, the project is now governed by a community of maintainers, with Junio Hamano leading the way for years. There isn't a corporate boardroom making decisions behind closed doors; instead, there are mailing lists and IRC channels where every change is debated openly. This level of transparency is exactly what makes me trust Git for my own security research and academic projects.

PartD —Open Source vs Proprietary (CriticalAnalysis)

Compare your chosen software against its closest proprietary alternative. Be specific and honest — do not assume open source is always better.

Dimension	Open Source	Proprietary Alternative
Cost	Free (Freedom & Price)	Expensive per-user licensing
Security (who can audit the code?)	Crowdsourced. Anyone can read the source code. Vulnerabilities are found and patched by a global community.	Vendor-Locked. Only the company (Perforce) can see the code. You have to trust their internal security team.
Support and reliability	Community-based. Help comes from forums, StackOverflow, and IRC. No "guaranteed" response time.	Contractual. Paid support with Service Level Agreements (SLAs). You can call a helpdesk for 24/7 fixes.

Freedom to modify	Absolute**.** If I do not like a feature, I can fork the code and change it. I own the tool.	None**.** The software is a "black box." Modifying it would violate the EULA (End User License Agreement).
Community vs corporate control	Meritocracy**.** The direction is set by the most active developers and maintainers globally. No one can "kill" the project.	Monopoly. The roadmap is decided by the company's profit goals. They can stop supporting a version whenever they want.
Your overall verdict	Highly Recommended**.** It is the industry standard for a reason: it is fast, secure, and gives the developer total ownership of their history.	Niche Recommendation**.** Only worth it for massive corporate teams with huge binary files (like 4K video assets) who need paid support.

Fill in the table as part of your report, then write a two-paragraph conclusion: given everything you have learned, would you recommend your chosen software for a real-world deployment? Would you contribute to it?

Answer: After digging deep into how Git really works, I would absolutely recommend it for almost any real-world deployment, especially for modern software teams. Its distributed and open nature means every developer has a full backup of the project history, which is a massive safety net that centralized systems often lack. While perforce might be better for handling massive 100GB binary files in AAA game dev, for 99% of coding projects, Git is speed and the size of its ecosystem make it the only possible available logical choice. It is not just about the zero price tag; it is about the fact that if I run into a bug at 2:00 AM, a thousand other people have already found the fix and posted it online Regarding contribution, I definitely see myself giving back to the Git ecosystem, though perhaps not by writing core C code for the kernel just yet. I'm more interested in contributing through documentation or by creating better "hooks" and automation scripts to help teams follow better branching workflows. Open source only stays alive if people do not just "take and use" but also "give" and after seeing how much I have relied on Git during my time at VIT-Bhopal, I have come to my senses that it really is very helpful.

4.Shell Script Tasks

You will write five shell scripts. Each script must be submitted as a plain .sh file and documented inside your report with: the script code, a screenshot of it running, and a paragraph explaining what it does and which shell scripting concepts it uses.

Important	Scripts must run on a real Linux system. They must include comments explaining each section. Uncommented scripts will lose marks.

Script1 —System IdentityReport (Unit 1&2)

Write a script that introduces the Linux system like a welcome screen. It should display:

- The Linux distribution name and kernel version
- The current logged-in user and their home directory
- System uptime and current date/time
- A message stating which open-source license covers the OS

Concepts to use: variables, echo, command substitution ($()), basic output formatting.

Starter structure

```
#!/bin/bash
# Script 1: System Identity Report
# Author: Ayush Mishra [cite: 103]

STUDENT_NAME="Ayush Mishra"
SOFTWARE_CHOICE="Git"
KERNEL=$(uname -r)
DISTRO=$(cat /etc/os-release | grep "PRETTY_NAME" | cut -d'"' -f2)

echo "================================"
echo "  $SOFTWARE_CHOICE Audit Report  "
echo "================================"
echo "Student: $STUDENT_NAME"
echo "Distro : $DISTRO"
echo "Kernel : $KERNEL"
echo "Uptime : $(uptime -p)"
echo "License: This OS is GPL protected."
```

Answer:
For my first script, I wanted to create a personalized welcome screen that acts as a quick 'System Identity Card.' When I run this on my Kali Linux terminal, it instantly pulls the most critical environment data so I do not have to hunt through different

commands. It tells me exactly which kernel version I'm working with and reminds me of the legal framework (GPL v2) that makes this entire OS possible. I found this particularly useful for my audit because knowing the uptime and the current user context is the first step in any security reconnaissance or system check.

I relied heavily on Command Substitution using the $(...) syntax, which allowed me to take the output of system commands like uname or hostnamectl and store them in Variables for later use. I also used the cut command to clean up the output strings so the report looks professional and readable. Finally, I used the echo(print) command to format the visual layout, creating a clear structure for the data. This taught me how to bridge the gap between raw system data and a user-friendly interface.

```
┌──(xcenz㉿s7-pm)-[~]
└─$ ./script1.sh
================================
  Git Audit Report
================================
Student: Ayush Mishra
Distro : Kali GNU/Linux Rolling
Kernel : 6.6.87.2-microsoft-standard-WSL2
Uptime : up 0 minutes
License: This OS is GPL protected.

┌──(xcenz㉿s7-pm)-[~]
└─$
```

Script 2 — FOSS Package Inspector (Unit 2)

Writea script that checks whetheryour chosensoftwareis installed on the system, finds its version, and uses a case statement to print a short description of its purpose based on the package name.

Concepts to use: if-then-else, case statement, rpm -qi or dpkg -l, pipe with grep.

Starter structure

```
#!/bin/bash

#
=====================================================
====
# Script Name: package_inspector.sh
# Purpose: To audit the presence of Git on my Kali
system.
# Concepts: Dpkg, Case statements, Grep, and If-Else
logic.
#
=====================================================
====

# I'm defining my target software here
SOFTWARE="git"

echo "--- Starting Package Audit for: $SOFTWARE ---"

# 1. Checking if the package is installed using dpkg
(standard for Kali/Debian)
# I'm piping the output to grep to see if the status
is 'installed'
if dpkg -l | grep -q "^ii  $SOFTWARE "; then

    # 2. If found, let's grab the specific version
string
    VERSION=$(git --version | awk '{print $3}')
    echo "[FOUND] $SOFTWARE version $VERSION is
correctly installed."
    # 3. Using a CASE statement to describe the
software's role
    # This makes the script easily expandable for
other tools later
    case $SOFTWARE in
        "git")
            echo "Purpose: This is a Distributed
Version Control System (DVCS)."
            echo "It handles source code integrity and
collaboration."
            ;;
        "apache2")
            echo "Purpose: This is a robust Open
Source HTTP Web Server."
            ;;
        *)
            echo "Purpose: This is an unidentified
Open Source utility."
            ;;
    esac
else
    # 4. Error handling if I haven't installed it yet
    echo "[MISSING] $SOFTWARE was not found on this
system."
    echo "Action: Run 'sudo apt update && sudo apt
install $SOFTWARE' to fix this."
fi

echo "--- Audit Complete ---"
```

Answer: For my second script, I wanted to move beyond just 'checking' if a file exists and actually query the system's package manager. Since I am working on Kali Linux, I used the dpkg -l command. I paired this with a pipe and grep to search for the specific 'installed' status. The highlight of this script for me was the Case Statement. Instead of using a messy and wired string of 'if' statements to describe what the software does, the case block provides a clean, professional way to categorize the tool, case block provides gradual working on decision making. When I ran this, it correctly identified Git as a Distributed Version Control System and pulled the exact version number, which is essential for documenting my lab environment.

This script was a great exercise in Conditional Logic. I used an If-Then-Else block to split the script's behaviour based on whether the software was found or missing. I also practiced Command Output Filtering by using awk to isolate just the version number from the standard git output. Finally, the Case Statement allowed me to add a layer of human-readable description to the technical data. Learning how to pipe the results of one command (dpkg) into another (grep) showed me how powerful the Linux 'Lego-block' philosophy of small, connected tools really is.

```
┌──(xcenz㉿s7-pm)-[~]
└─$ ./script2.sh
--- Starting Package Audit for: git ---
[FOUND] git version 2.51.0 is correctly installed.
Purpose: This is a Distributed Version Control System (DVCS).
It handles source code integrity and collaboration.
--- Audit Complete ---

┌──(xcenz㉿s7-pm)-[~]
└─$
```

Script 3 — Disk and Permission Auditor (Unit 2)

Write a script that loops through a list ofimportant system directories and reports: how much space each uses, and the owner and permissions of the directory. Use a for loop.

Concepts to use: for loop, df, ls -ld, awk or cut to extract fields.

Starterstructure

```
#!/bin/bash

# ==========================================================
# Script Name: log_analyzer.sh
# Purpose: Search for security or system errors in logs.
# Concepts: User input (read), Grep, Word Count (wc), and Tail.
# ==========================================================
# I'm using a common log file on Kali/Debian systems
LOG_FILE="/var/log/syslog"

echo "----------------------------------------------------------"
echo "LOG SECURITY ANALYZER"
echo "----------------------------------------------------------"
# 1. Taking user input to decide what we are hunting for
echo -n "Enter the keyword to search for (e.g., ERROR, FAILED, CRITICAL): "
read KEYWORD

echo "Searching for '$KEYWORD' in $LOG_FILE..."

# 2. Counting the occurrences using grep and wc -l
COUNT=$(grep -i "$KEYWORD" "$LOG_FILE" | wc -l)
if [ "$COUNT" -gt 0 ]; then
    echo "[!] ALERT: Found $COUNT occurrences of '$KEYWORD'."
    echo "----------------------------------------------------"
    echo "Last 5 entries found:"

    # 3. Showing only the most recent entries using tail
    grep -i "$KEYWORD" "$LOG_FILE" | tail -n 5
else
    echo "[OK] No entries for '$KEYWORD' were found."
fi

echo "----------------------------------------------------"
```

Answer: For my third script, I wanted to build a tool that could help me perform a 'health check' on the system's file structure. I created a For Loop, which is a conditional loop, that cycles through a specific list of directories: /etc, /var/log, /usr/bin, and my own home folder. For each path, the script calculates how much disk space is being consumed and, more importantly for a security audit, identifies who owns the directory and what the permission bits are set to. This helps me verify that sensitive areas like /etc aren't accidentally left open for anyone to modify.

This script was my first real deep-dive into Looping Structures. By using a for loop, I was able to automate a repetitive task that would have taken four separate commands to do manually. I also got much more comfortable with Output Parsing using awk. I used awk to 'slice' the output of ls ld so I could isolate just the permissions and the owner’s name without all the extra metadata. At last finally, I used the -e flag with echo to create a tabbed table layout, which makes the data much easier to read at a glance during a live audit.

```
┌──(xcenz㉿s7-pm)-[~]
└─$ ./script3.sh
-------------------------------------------------------
SYSTEM DIRECTORY AUDIT: SPACE & PERMISSIONS
-------------------------------------------------------
PERMISSIONS     OWNER   SIZE    DIRECTORY
drwxr-xr-x      root    8.3M    /etc
drwxr-xr-x      root    555M    /var/log
drwxr-xr-x      root    380M    /usr/bin
drwx------      xcenz   11G     /home/xcenz
-------------------------------------------------------
Audit complete. Review permissions for any 'world-writable' risks.

┌──(xcenz㉿s7-pm)-[~]
└─$ |
```

Script 4 — Log File Analyzer (Units 2 & 5)

Writea scriptthatreads alog file line byline,counts how many lines contain a keyword (like ERROR or WARNING), and prints a summary. Use a while-read loop and an if-then inside it.

Concepts to use: while read loop, if-then, counter variables, command-line arguments ($1).

Starter structure

```
#!/bin/bash

# Script 4: Log File Analyzer
# Usage: ./log_analyzer.sh <logfile> [keyword]

LOGFILE=$1
# Default keyword is 'error' if the second argument is missing
KEYWORD=${2:-"error"}
COUNT=0

# Check if the log file exists and is a regular file
if [ ! -f "$LOGFILE" ]; then
    echo "Error: File '$LOGFILE' not found."
    exit 1
fi
     # TODO: Do-while style retry if the file is empty
# In Bash, we check the file size (-s) to see if it's empty
while [ ! -s "$LOGFILE" ]; do
      echo "File is empty. Retrying in 2 seconds..."
      sleep 2
done
echo "Analyzing $LOGFILE for keyword: '$KEYWORD'..."

# Main loop: reads the file line by line
while IFS= read -r LINE; do
    # Check if the line contains the keyword (case-insensitive -i)
    if echo "$LINE" | grep -iq "$KEYWORD"; then
        COUNT=$((COUNT + 1))
    fi
done < "$LOGFILE"
# Print the summary
echo "-------------------------------------"
echo "Summary: Keyword '$KEYWORD' found $COUNT times in $LOGFILE"

# TODO: Print the last 5 matching lines using tail + grep
echo "Last 5 matching entries:"
grep -i "$KEYWORD" "$LOGFILE" | tail -n 5
echo "-------------------------------------"
```

Answer: For my fourth script, I wanted to build a more granular log auditor. While a simple grep command can find words quickly, I used a while-read loop to process the log file one line at a

time. This is I used a much more powerful approach for a security researcher because it allows the script to perform logic on every single line it reads. I designed the script to take a filename as a commandline argument ($1), making it reusable for any log file on my Kali system, such as /var/log/bootstrap.log or a custom application log.
This script was my introduction to Stream Redirection and Counters. By using done (<) "$LOG_FILE", I redirected the file's content into the loop. Inside, I used a counter variable (MATCH_COUNT) that increments every time the if-then statement finds a match. I also practiced using Positional Parameters ($1), which taught me how to make my scripts interactive from the moment they are launched in the terminal. This 'line-by-line' logic is the foundation for how many professional Intrusion Detection Systems (IDS) actually parse live data.

```
┌──(xcenz㉿s7-pm)-[~]
└─$ ./script4.sh my_app.log "WARNING"
Analyzing my_app.log for keyword: 'WARNING'...
----------------------------------------
Summary: Keyword 'WARNING' found 2 times in my_app.log
Last 5 matching entries:
[WARNING] Disk space low
[WARNING] High memory usage
----------------------------------------
```

Script5 —The Open Source Manifesto Generator(Unit5)

This is the most creative script. Write a script that generates a personalized open source philosophy statement by asking the user three questions interactively, then composing a short paragraph using their answers and saving it to a .txt file.

Concepts to use: read for user input, string concatenation, writing to a file with >, date command, aliases concept demonstrated via a comment.

Starter structure

```bash
#!/bin/bash
# Unit 5: Open Source Manifesto Generator
# Author: Sankalp Tripathi (24BCY10005)
# Note: I could set an alias for this like: alias gen='./manifesto_gen.sh'
echo "--- Welcome to the Manifesto Builder ---"
echo "Please answer a few questions to build your philosophy statement."
echo ""
# 1. Using 'read' for interactive user input
read -p "What is your name? " user_name
read -p "What is your favorite open source tool (e.g., Linux, Git)? " fav_tool
read -p "In one word, why do you like Open Source? " reason
# 2. Getting the current date
current_date=$(date +%D)
# 3. String concatenation to build the manifesto paragraph
# We keep the language simple and direct
line1="I, $user_name, believe that software should be open to everyone."
line2="My journey started with $fav_tool, which taught me the power of community.
"
line3="To me, the core of open source is $reason."
line4="Generated on: $current_date"
# Combine them into a single block
manifesto_text="$line1 $line2 $line3 $line4"
# 4. Writing to a file using '>'
# This creates a new file or overwrites an existing one
echo "$manifesto_text" > my_manifesto.txt
echo ""
echo "Done! Your personalized manifesto has been saved to 'my_manifesto.txt'."
echo "----------------------------------------"
```

Answer: For my fifth script, I wanted to try something more interactive. While Script 4 was all about scanning logs, this one actually talks to the user to build a personalized document. It was a good

way to get comfortable with the read command it is interesting how the script pauses and waits for the operator to type in strings like their name or a favourite tool before moving on.
I spent a lot of time working on the string concatenation here. It was a bit of a puzzle to take those different variables and stitch them together into a paragraph that actually made sense. I also made sure to pull in the date command so the final file feels more like an official record, which is something I see a lot in professional documentation.
The biggest takeaway for me was using the (>) operator for output redirection. Usually, I just echo things to the terminal, but sending that data straight into a .txt file showed me how these scripts can be used to generate reports or config files on the fly. I also threw in a comment about aliases to show how I'd make this tool easier to launch in a real environment. This project really helped me see that Bash isn't just for fixing broken systems it is actually a pretty creative way to handle file management and user data.

```
┌──(xcenz㉿s7-pm)-[~]
└─$ ./script5.sh
--- Welcome to the Manifesto Builder ---
Please answer a few questions to build your philosophy statement.

What is your name? Ayush Mishra
What is your favorite open source tool (e.g., Linux, Git)? Git
In one word, why do you like Open Source? free

Done! Your personalized manifesto has been saved to 'my_manifesto.txt'.
----------------------------------------
```

5.Project Timeline

This timelineis a guide, not a rigid rule. Some students will move faster on the technical parts; others will spend more time on the philosophical sections. Adjust based on your strengths — but do not leave the scripts to the final stretch.

Day	Focus	Tasks	Deliverable
1	**Setup**	Choose software. Research its origin story. Read the license text. Set up your Linux environment (VM or lab system). Install your chosen software.	*Choice confirmed*
2	**Philosophy**	Write Part A1 — the origin story. Take notes on what problem the software solved and for whom.	*Draft A1*
3	**License & Ethics**	Write Part A2 — license analysis. Write Part A3 — ethical reflection. These sections need your own thinking, not summaries.	*Draft A2 + A3*
4	**Linux Footprint**	Install software on Linux. Document directories, user, permissions. Write Part B.	*Draft Part B + screenshots*
5	**Scripts 1 & 2**	Write and test Script 1 (System Identity) and Script 2 (Package Inspector). Add them to your report with explanations.	*Scripts 1–2 working*
6	**Scripts 3 & 4**	Write and test Script 3 (Disk Auditor) and Script 4 (Log Analyzer). Test with a real log file.	*Scripts 3–4 working*
7	**Ecosystem**	Write Part C — FOSS ecosystem map. Research dependencies and community.	*Draft Part C*
8	**Comparison & Script 5**	Write Part D — comparison table and verdict. Write and test Script 5 (Manifesto Generator).	*Draft Part D + Script 5*
9	**Report polish**	Compile full report. Write introduction and conclusion. Add screenshots. Check page count (12–16 pages).	*Complete draft*
10	**Submission**	Final review. Submit report + 5 .sh files. Be ready to explain any script in a short viva if asked.	*Final submission*

6. Evaluation Rubric

Total marks: 100. The reportcarries 60 marks and the shell scripts carry 40 marks. There is no viva unless the evaluator suspects the work is not original — in which case any student may be called for a brief oral explanation.

Report (60 marks)

#	Component	Full marks	Passing (50%)	Notes
1	Origin story — depth, original research, storytelling	12	6	*Part A1*
2	License analysis — accuracy and understanding of freedoms	12	6	*Part A2*
3	Ethical reflection — quality of argument and original thinking	10	5	*Part A3*
4	Linux footprint — technical accuracy and screenshots	10	5	*Part B*
5	Ecosystem and LAMP connection	8	4	*Part C*
6	Open source vs proprietary comparison	8	4	*Part D*

ShellScripts(40marks—8marks each)

Each script is marked on three criteria:

Criterion	Marks	What we look for
Correctness — does it run without errors?	4	*Script executes on Linux and produces expected output*
Concepts — does it use the right shell constructs?	2	*Loops, if-then, case, variables used appropriately for the task*
Comments and documentation	2	*Every non-obvious line has a comment; script purpose is clear*

Academic integrity	All written sections must be in your own words. Code that is copy-pasted from the internet without understanding will be zero-marked. Running a script and explaining what it does in a conversation is considered a passing demonstration of understanding.

7.Reference Resources

Usetheseasstartingpoints,notassources to copy from. The report must be in your own words.

On opensourcephilosophy

- GNU Project — 'The Free Software Definition': gnu.org/philosophy/free-sw.html • Open Source Initiative — 'The Open Source Definition': opensource.org/osd
- Eric S. Raymond — 'The Cathedral and the Bazaar': catb.org/~esr/writings/cathedral-bazaar
- Linus Torvalds — 'Just for Fun' (autobiography, available at most libraries)

OnLinuxandshellscripting

- The Linux Command Line by William Shotts — freely available at linuxcommand.org
- GNU Bash Manual: gnu.org/software/bash/manual
- Linux man pages: man.cx (online) or type 'man <command>' in your terminal

For finding license texts

- SPDX License List: spdx.org/licenses
- Choose a License: choosealicense.com
- Your software's official repository on GitHub or GitLab — the LICENSE file is always there

A closing thought

Every tool you will use in your career — the editor, the compiler, the server, the database — was shaped by people who chose to build in the open and share their work freely. Understanding why they made that choice, and what it means for the world, is not a footnote to your technical education. It is the foundation of it.

8.Submission Requirements

All submissions are made through the VITyarthi portal. You must submit three things: a public GitHub repository link, a README.md inside the repository, and a project report PDF. All three are compulsory. Incomplete submissions will not be evaluated. **What to submit on the VITyarthi portal**

Item	Details	Compulsory?
GitHub repo link	The repository must be public. It must contain all five .sh script files and a README.md. Name the repo: oss-audit-[rollnumber]	**Yes**
README.md	Must be inside the GitHub repository. Must include: student name and roll number, chosen software, description of each script, step-by-step instructions to run each script on Linux, and any dependencies required.	**Yes**
Project report PDF	Upload directly on the VITyarthi portal. Must follow the structure in Section 3. Must be 12 to 16 pages. Must be your own writing.	**Yes**

On the VITyarthi portal submission form, paste the GitHub repository URL, confirm the README is present in the repository, and upload the project report PDF. All three must be submitted together in a single submission — partial submissions will not be accepted.

Evaluation pipeline

Every submission goes through an automated evaluation pipeline before it reaches an evaluator. Be aware of the following checks that run on every submitted report and script set.

Check	What it detects	Consequence
Plagiarism detection	Report text copied from websites, Wikipedia, other students' submissions, or any published source without attribution. Cross-submission similarity between enrolled students is also flagged.	Zero on the full report. Referred to disciplinary committee if similarity exceeds 40%.

AI generation detection	Report sections generated by AI tools such as ChatGPT, Gemini, Claude, or similar. The pipeline checks for linguistic patterns, sentence uniformity, and structural signatures common to AI-generated text.	Flagged submissions are called for a mandatory viva. If the student cannot explain the content, zero marks are awarded.

A note on AI tools: you may use AI to understand concepts, debug scripts, or get explanations. You may not use AI to write your report. The distinction is the same as using a calculator versus asking someone else to sit your exam. The report must reflect your thinking, your words, and your conclusions.